Middle East Technical University Department of Computer Engineering

CENG 351

Data Management and File Structures

SAMPLE

In-Class Assignment 4 – B+ Tree

Duration: 80 Minutes

REMARK-1: For questions 1 and 2 (B+ tree insertion and deletion), apply algorithms as shown in the course slide sets. That is:

- In case of overflow in any node, apply node **split** as shown in the lectures (Note that, we **don't** apply re-distribution for resolving overflows, although such a possibility is mentioned in the textbook).
- In case of underflow in any node, first try **redistribution** and if this is not applicable, apply node merge, as shown in the lectures.

REMARK-2: To facilitate drawing the trees, we did not show the empty places in the nodes and asterisks on top of data entries in the leaf nodes; as we usually did in the course slide sets.

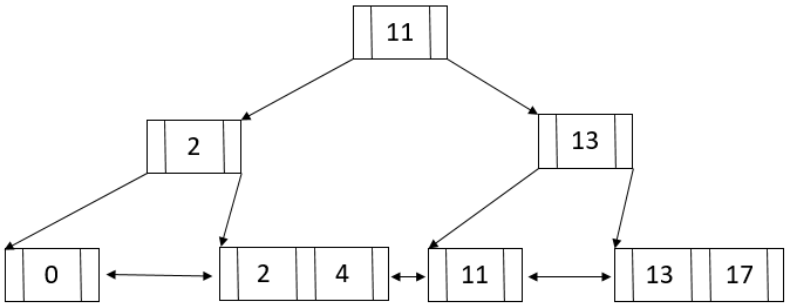
1. B+ Tree Insertion

The following keys are given:

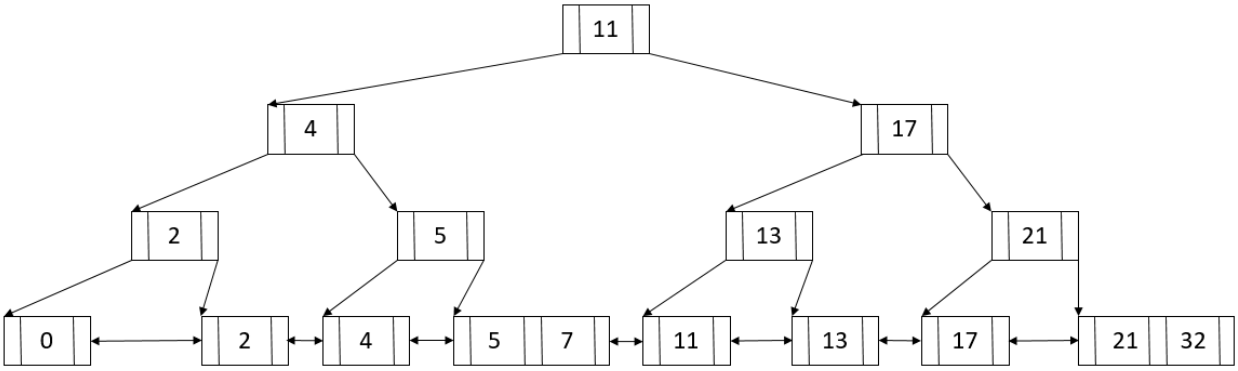
[11, 17, 4, 2, 0, 13, 21, 32, 7, 5, 9, 18, 10, 6, 12]

You are going to create two different B+ trees by succesive insertions of the given keys. You will create a B+ tree of order 1 for (a) and (b), and a B+ tree of order 2 for (c) and (d).

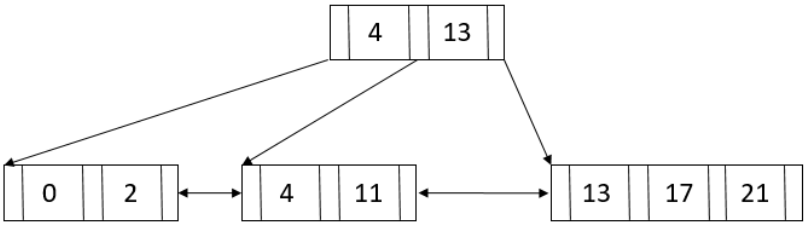
(a) Show the B+ tree of **order 1** after the insertion of **13**. (10 Points)



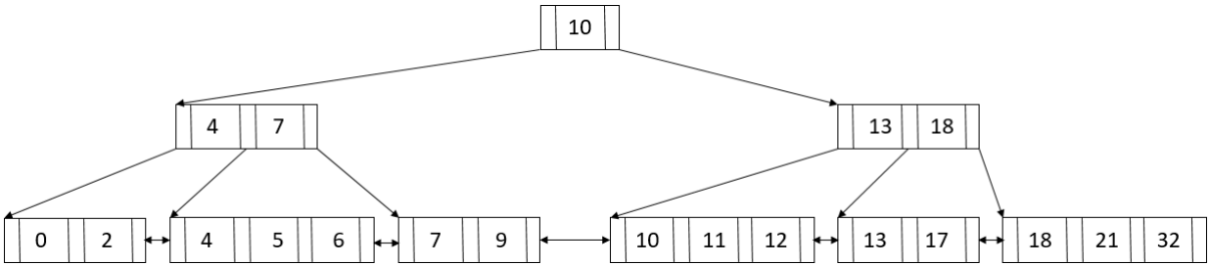
(b) Show the B+ tree of **order 1** after the insertion of **5**. (10 Points)



(c) Show the B+ tree of **order 2** after the insertion of **21**. (10 Points)



(d) Show the B+ tree of **order 2** after the insertion of **12**. (10 Points)

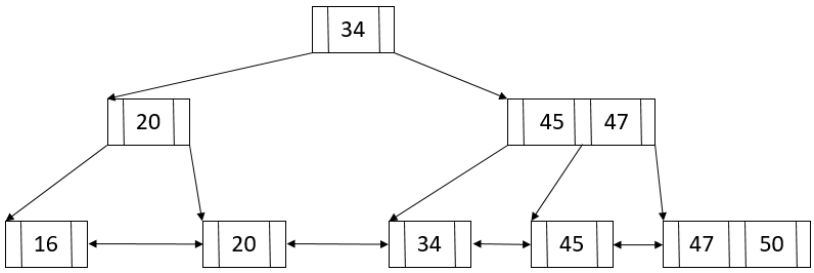


2. B+ Tree Deletion

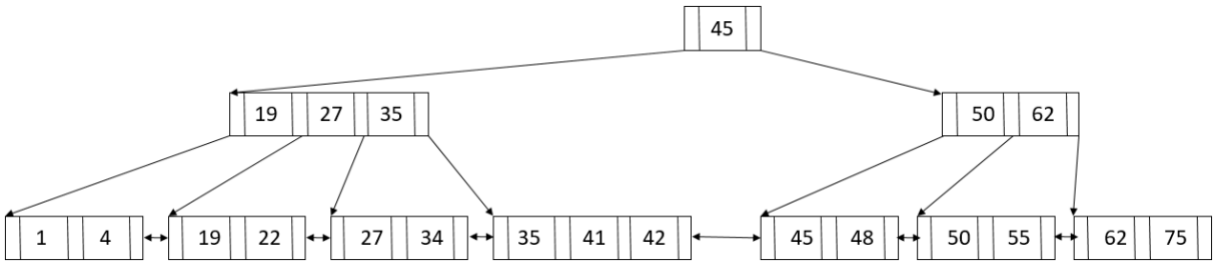
In this part, you are going to apply delete operations on different B+ trees. For (a) and (b) you will delete from order 1 B+ tree X, for (c) and (d) from order 2 B+ tree Y.

Note: They are all different cases, solve separately. In other words, all deletion operations use the initial state of X and Y, not the changed ones.

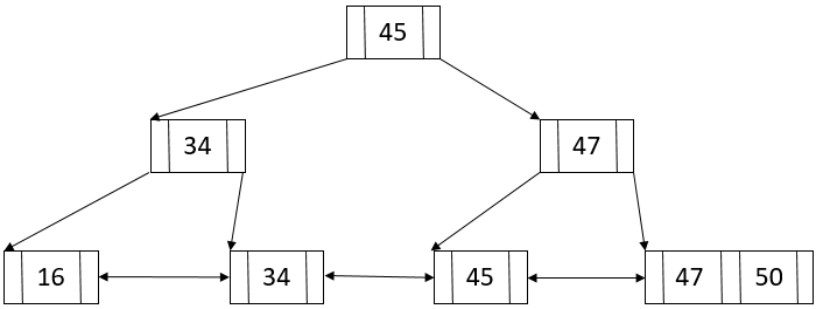
X:



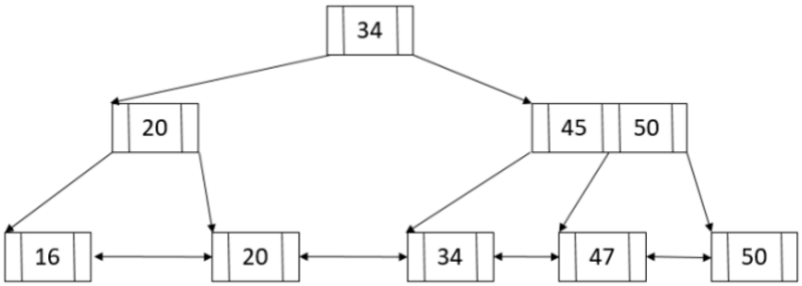
Y:



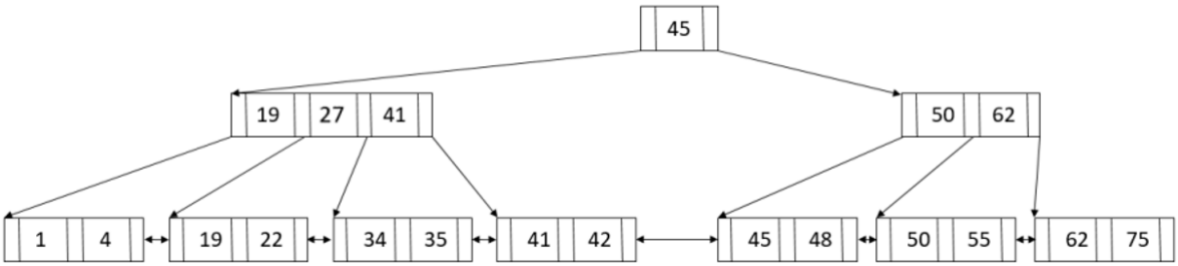
(a) Delete **20** from X (10 Points)



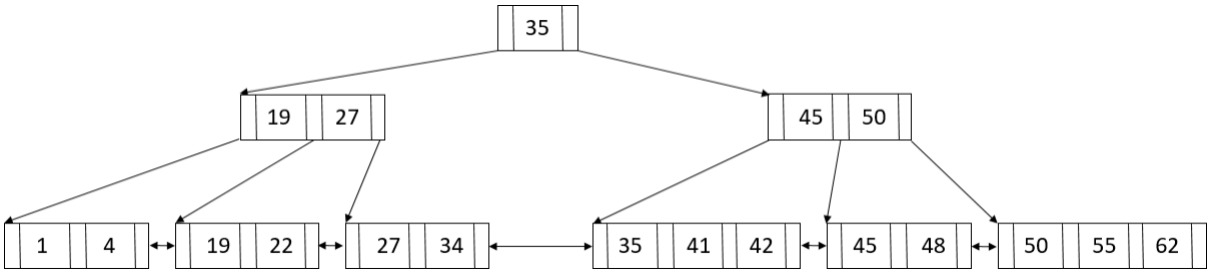
(b) Delete **45** from X (10 Points)



(c) Delete **27** from Y (10 Points)



(d) Delete **75** from Y (10 Points)



3. PhoneBook

We plan to create a **secondary B+ tree index** on key field phoneKey. The size of phoneKey is 16 bytes, the size of a pointer is 4 bytes, and the block size of the disk is 400 bytes. The number of records to be stored is 200,000.

(a) What should be the order of the tree? (i.e. maximum possible order) (5 Points)

16*n + 4*(n+1) = 400
20n=396
n=19 → 2d ?= 19

$order(d)$: ⌊19 / 2 ⌋ = 9 (and hence, 2d is 18)

(b) How many leaf nodes are required for indexing all the records if the average fill factor is 80%? (Assume that we keep a key and pointer for each data record at the leaf nodes, and each leaf node is one disk block. You can ignore the two-way pointers between the leaf nodes.) What would be the number of levels (including the leaf level) in such a B+ tree? (5 Points)

Max no of keys in an index node can be 2d; so with 80% fill factor:

Number of keys per index node is: ⌊ 18 * 80% ⌋ = 14 and hence, fanout (F) = 15

Number of leaf nodes: ⌈200000/14⌉=14286
(Recall that there would be F-1 data entries per leaf)

$$Number\ of\ \ levels: \lceil log_{15}(14286) \rceil + 1\ (leaf\ level) = 4 + 1$$

(c) What is the maximum possible number of levels (including the leaf level)? Take the answer of (a) as the order. Show your work. (5 Points)

Tree will have max levels when the nodes have minimum occupancy, i.e., with d entries. In part (a), d = 9, so F = 10 in the internal nodes.

$$Number\ of\ leaf\ nodes: \lceil 200000/9 \quad \rceil = 22223$$
(Recall that there would be F-1 data entries per leaf)

$$Number\ of\ \ levels: \lceil log_{10}(22223) \rceil + 1\ (leaf\ level) = 5 + 1 = 6$$

(d) What is the minimum possible number of levels (including the leaf level)? Take the answer of (a) as the order. Show your work. (5 Points)

Tree will have min levels when the nodes have maximum occupancy, i.e., with 2d entries. In part (a), d= 9 and 2d = 18, so F = 19 in the internal nodes.

$$Number\ of\ leaf\ nodes: \lceil 200000/18 \quad \rceil = 11112$$
(Recall that there would be F-1 data entries per leaf)

$$Number\ of\ \ levels: \lceil log_{19}(11112) \rceil + 1\ (leaf\ level) = 4\ + 1 = 5$$